Adapting Bias Evaluation to Domain Contexts using Generative Models

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TL;DR

NLP models are used across many domains, but social bias datasets often ignore this variation. We propose a low-cost, LLM-based method to adapt template datasets to the target domain, yielding more realistic estimates.

Introduction

- There are two common **approaches** to develop datasets to **measure social bias** in NLP systems.
  - **Template-based datasets:** Manually written sentences (e.g., “[X] feels [emotion]”).
    - Scalable.
    - Synthetic; often lexically mismatched with real text.
  - **Naturally Occurring Examples (NOEs):** Sentences extracted from real domains (e.g., Wikipedia).
    - Realistic.
    - Costly to collect/annotate; uneven coverage across groups.

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- **Problem**: Neither approach alone can estimate bias in a scalable and adaptable way across diverse domains.
- This is relevant, as NLP models are **deployed across many domains**, and dataset–domain **mismatch can misestimate bias**.
- We introduce a method that **converts template datasets into domain-specific variants,** improving realism while retaining scalability.

Methodology

Given a template-based dataset T and a domain $\mathcal{D}$, we create a domain-adapted set $T_{\mathcal{D}}$, by asking an LLM to rewrite each template $t \in T$ as a text from $\mathcal{D}$, using **n** random in-domain examples sampled from $\mathcal{D}$.

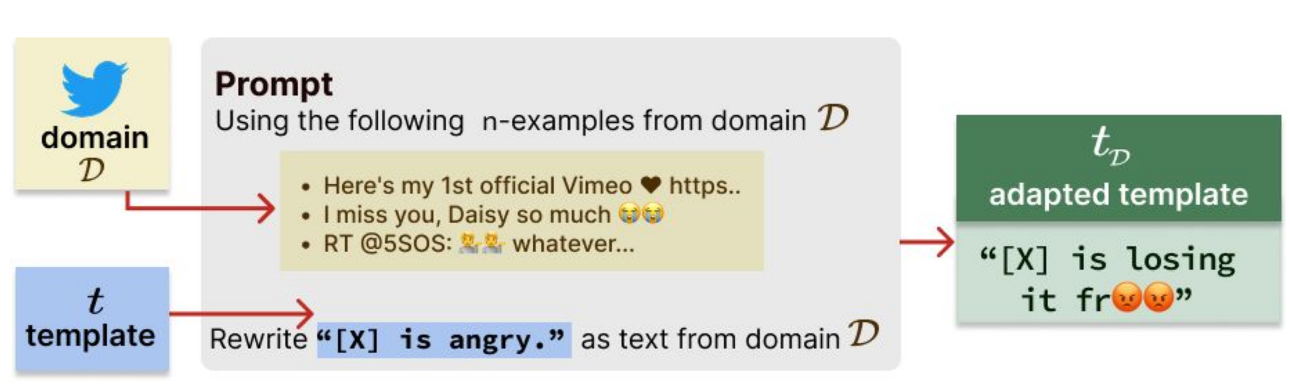

Fig. 1. Template adaptation process.

To test the effectiveness of $T_{\mathcal{D}}$:

- We build a reference dataset N by selecting sentences from the domain that contain **named entities**.
- We **hypothesize** that the bias measurement obtain with N will be **closer** to the one obtained with $T_{\mathcal{D}}$ than the one obtained with T, when adapted templates improve realism.

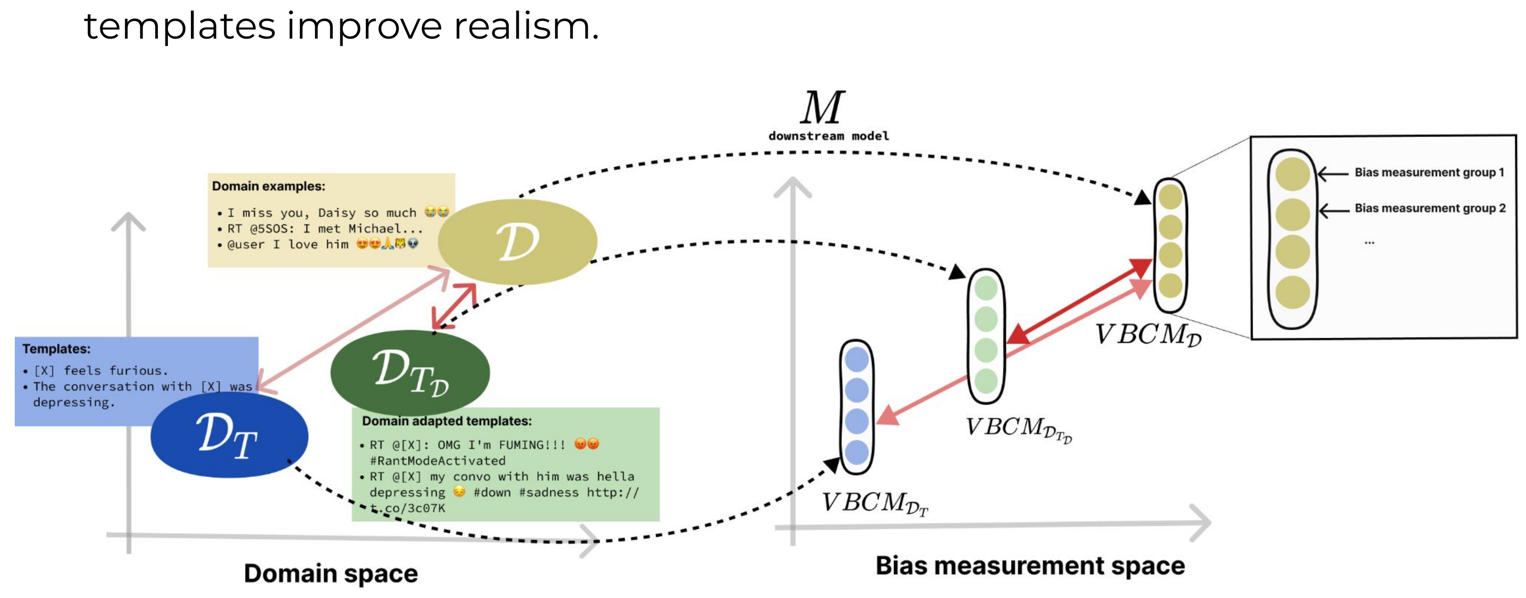

Fig. 2: Methodology to evaluate effectiveness template adaptation.

- The bias is **estimated** through the Vector Background Comparison Metric ($VBCM$) which is a vector where each ***i***-th component shows the bias of group ***i*** with respect a background group.
- We evaluate **closeness** of VBCM using two metrics: mean absolute error (**MAE**) and **Pearson** correlation (ρ) ; **↓MAE** and **↑Pearson** indicate more consistency.

Experimental Setup I

Sensitive attributes evaluated

- **Nationality bias**:
  - 38 **countries.**
  - **Names as proxies.** For each country, we use 50 popular names.
    - ➢ E.g: *“Alexander”* → *France, “Pedro”* → *Spain*
- **Gender bias**:
  - 4 groups: 2 **female groups** and 2 **male groups.**
  - **Name and nouns as proxies** for female/male groups.
    - ➢ Female/male noun. *E.g.: “the girl”* → *Female*
    - ➢ Female/male names. *E.g.; “Emma”* → *Female*

Experimental Setup II

Templates studied:

- *Equity Evaluation Corpus* (**EEC**) and *Identity Phrase Templates Test Set* (**IPTTS**) datasets.

EEC – Templates	**IPTTS – Templates**
I saw [PERSON] in the market.	*[PERSON] is an ugly nurse.*
[PERSON] feels happy.	*[PERSON] is a lovely fire fighter.*

Table 1: Examples of templates in the datasets.

Target Domains

- **Twitter** and **Wikipedia Talk Pages.** To select **NOEs** and examples for the adaptation process, we use the **EuroTweets for Twitter** and **Wikipedia Talks Pages test sets**.

Models to Generate Adaptations

- We use: **LLaMA-3 8B**, **LLaMA-3 70B**, and **Mixtral-8x7B** with **15-in domain examples**

Downstream Models and tasks

- **EEC:** evaluate **sentiment regression**.
- **IPTTS:** evaluate **toxicity classification**.

For each task., we assess **five** downstream models: **three fine-tuned** and **two off-the-shelf**

Results

Nationality bias

We compare **VBCM vectors of original templates and their adapted LLM version** across multiple models using **MAE** and **correlation**:

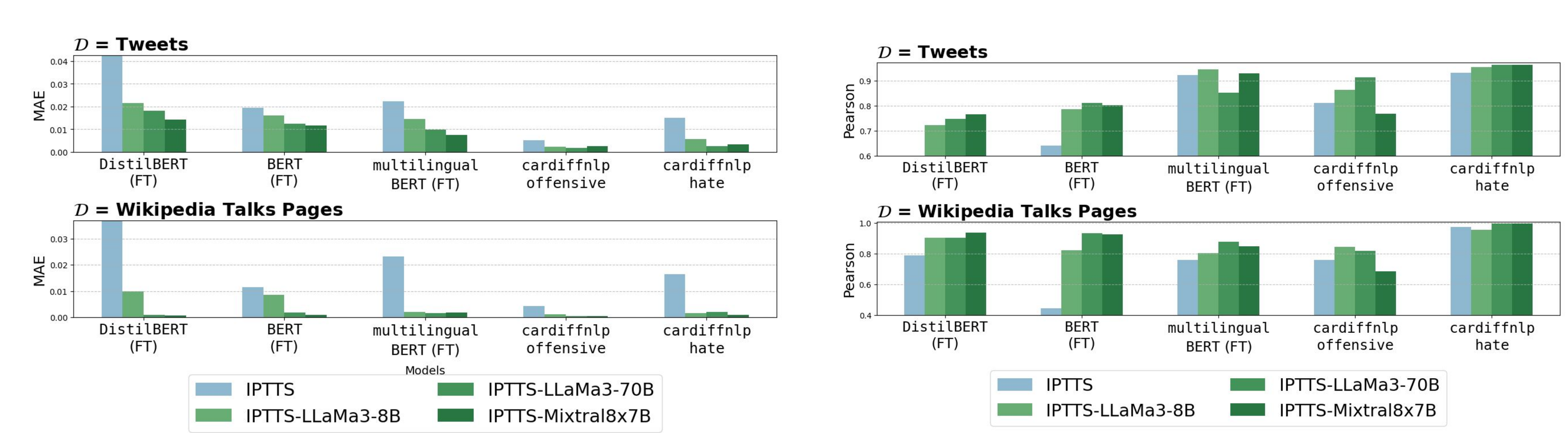

Fig. 3. MAE and Pearson correlation between bias vectors for templates and their adapted counterparts, measured against NOEs.

The correlation and error varies between downstream models, highlighting domain shift effects. In some cases, **original templates show low agreement** with NOEs—e.g., ***ρ* = 0.49** (EEC) and ***ρ* = 0.24** (IPTTS)—showing the limitations of curated datasets.
Across **domains**, **datasets**, and **models**, **adapted templates align more closely** with NOEs (**lower MAE**, **higher *ρ***).

Gender bias

We repeat the analysis (MAE, Pearson) across models and tasks, comparing **name-based** and **common-noun groups** to NOEs.

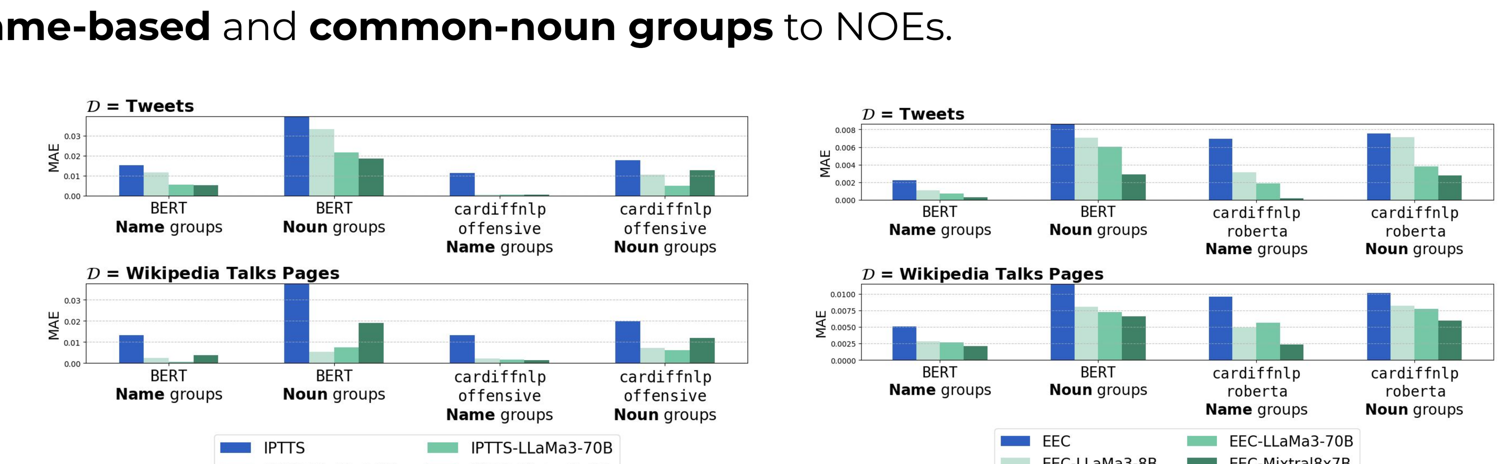

Fig. 4. MAE and Pearson correlation between bias vectors for templates and their adapted counterparts, measured against NOEs for the gender attribute.

The results are **similar** with the measurements of nationality bias. Furthermore, this proves that our method **produces good results** for different forms of **identity representation**.

Conclusions

- Template-based datasets **misestimate bias** compared to real corpus examples.
- We propose an **LLM-based domain adaptation** to address this issue; the method is **simple**, **low-cost**, and **adaptable** to any domain.
- Adapted templates **improve alignment with real text** across datasets, domains, and models, **enhances the realism** of bias measurement, a key limitation of current practice.